



Kisan Vaani

A multilingual voice advisory agent — the right farm guidance, to the right farmer, in their language, over their channel, every morning.

THE PROBLEM

Time-critical farm advice, lost in the last mile

Spray windows, irrigation calls, storm warnings — the guidance exists. Getting it to the right farmer, in time, in a form they can use, is where it breaks.



Literacy

Advisories go out as English or Hindi SMS — to people who farm, not read. Text-first fails the last mile.



Language

One national language can't reach 22+ regional mother tongues. The wrong language is the same as silence.

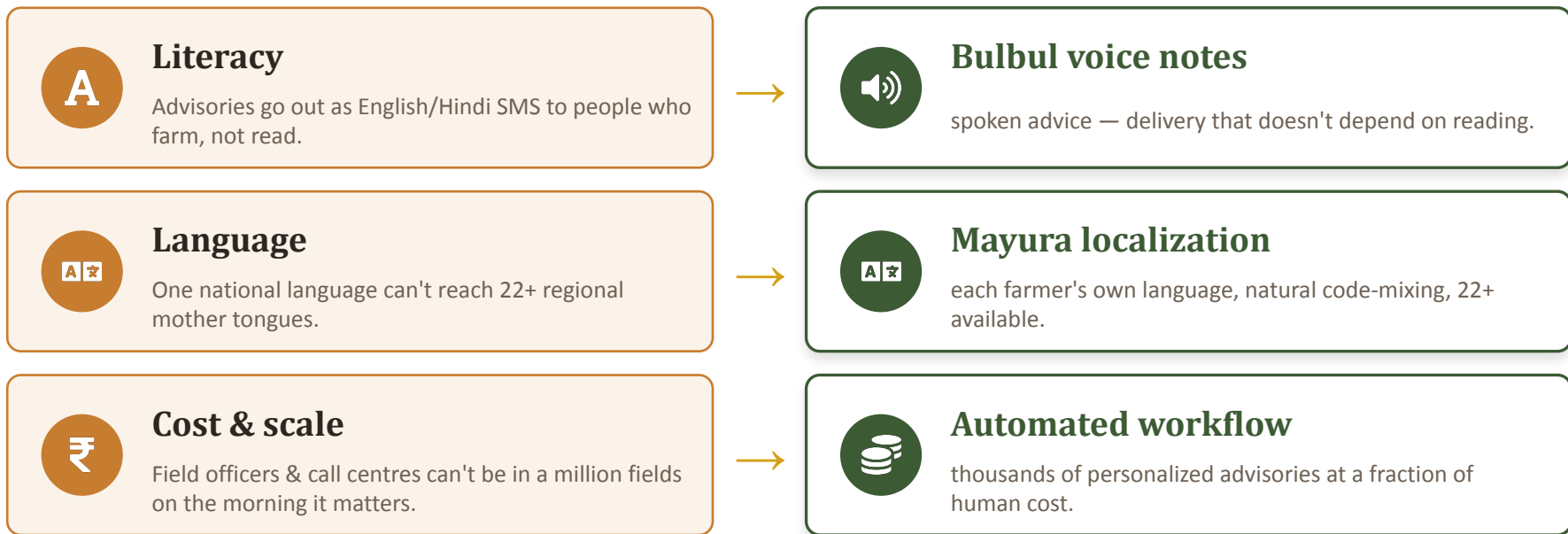


Cost & scale

Field officers and call-centre agents give great advice — but can't be in a million fields on the morning it matters.

THE CHALLENGE, AND HOW SARVAM ANSWERS IT

Every way advice reaches farmers today fails — voice-AI fixes each



Why voice, not better SMS: farmers are most comfortable in their mother tongue, fluent with voice notes, often on feature phones — the medium has to be voice, in their language.

THE STACK BEHIND IT

Four Sarvam APIs, one vendor

Each API does one job in the pipeline — all built for Indian languages, not retrofitted from a global model.



Chat Completions · sarvam-105b

Classifies the weather and selects the one agronomist-written rule. Rephrases expert advice — never invents it.



Translate · Mayura

Localizes each message into the farmer's language — Kannada and Marathi here, 22+ available.



Text-to-Speech · Bulbul v3

Speaks the advisory as a natural voice note — the reason the product is voice, not text.



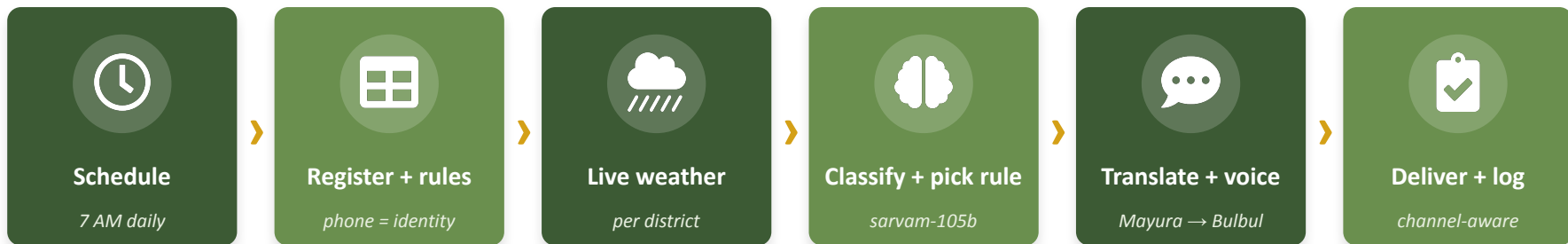
Speech-to-Text · Saaras v3

Lets a field officer feed the system by simply speaking one memo. Powers the campaign broadcast.

Why Sarvam over a generic stack: data residency in India · natural code-mixing · one vendor for LLM, translate, and voice.

THE WORKING MODEL

One rule table + live weather → the right voice note, every morning



Auditable by design — every message logs the rule ID it used; the AI only localizes expert content, it never authors agronomy. That is what makes automated farm advice safe.

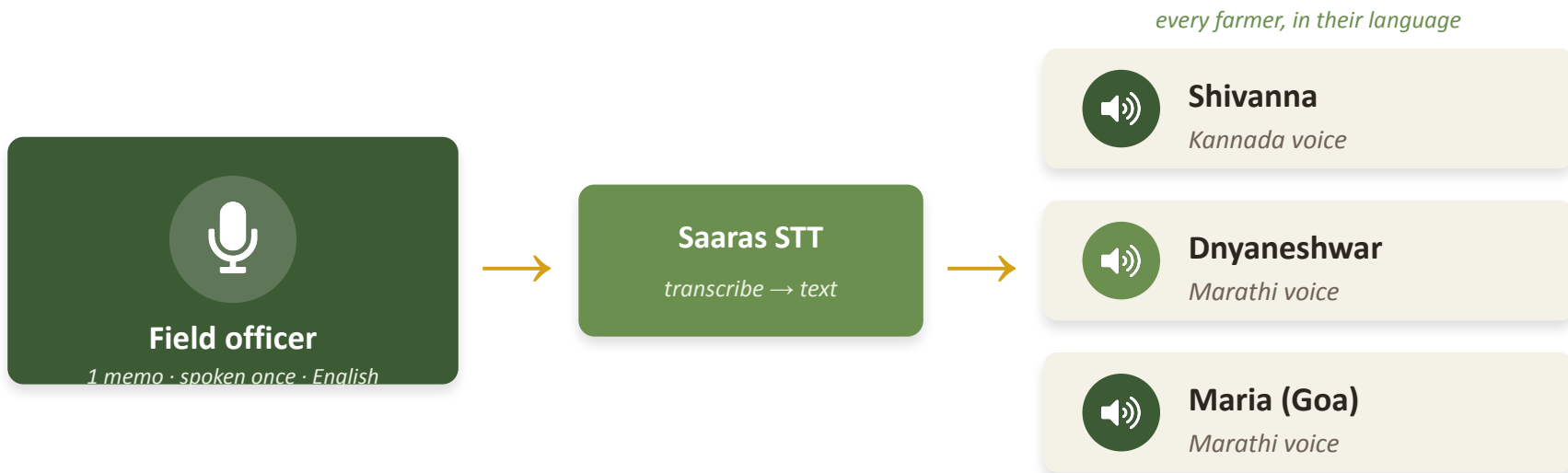
TWO WAYS TO DRIVE THE SAME PIPELINE

Scheduled daily advisory (this flow, per-farmer & weather-driven) + **On-demand campaign broadcast** (one spoken officer memo → every farmer, next slide).

CAMPAIGN MODE

One officer speaks once — every farmer, their language

The same translate + voice pipeline as the daily agent — driven by one human recording instead of a schedule.



The story: one spoken sentence reaches the entire farmer base — each person in the language they actually understand.

~64% lower cost per advisory — at a scale humans can't staff

10,00,000 farmers × 2 crops/year × 10 advisories/cycle = **2 crore advisories / year**

HUMAN CALL CENTRE

~₹50 cr/yr

≈ ₹25 / advisory · fully-loaded BPO ₹8–20/min

saves

~64%

KISAN VAANI (AUTOMATED)

~₹18 cr/yr

≈ ₹9 / advisory · APIs + delivery + retries + orchestration

≈ **₹32 crore saved every year** — and it scales flat when demand spikes; humans can't.



Proven pattern: Sarvam's voice stack already runs BFSI collections & loan-servicing for lenders like Tata Capital — the same outbound motion, under heavier compliance.

Automation also absorbs seasonal demand spikes and needs no per-language staffing. Cost is fully-loaded (APIs + delivery + retries + orchestration); measured Sarvam API spend across the build was ~₹31 total — under ₹1/advisory in compute. Sits within documented BFSI voice-AI savings of ~30–60%.

What the PoC deliberately isn't — and how it ships



Deliberate PoC scope

- Delivery simulated (demo inbox), not live channels
- Mock register, not a real CRM / FPO database
- Sample rules — real ones from the agronomy team
- No consent / DND management yet (TRAI)
- One-way broadcast — no farmer replies yet
- Konkani translate-only (no Bulbul voice yet)
- Weather per farmer — cache per district at scale

DAYS 0–30

Phase 1

Make delivery real

WhatsApp Business API pilot in 2 districts · real agronomy rule base · consent / DND capture · CRM replaces the sheet.

DAYS 31–60

Phase 2

Add the phone channel + measure

Telephony for feature phones (Exotel / Sarvam Samvaad) · IVR "press 1 to hear again" · per-district weather caching · listen-through analytics.

DAYS 61–90

Phase 3

Two-way + scale

Inbound farmer questions (Saaras STT + routing to agronomists) · 2 → 6+ languages · A/B voice advisory vs SMS control to prove the lift.

Phase 1 makes it real · Phase 2 makes it complete · Phase 3 makes it two-way and proven. No gap is a technical blocker — each is a scoped decision with a clear path.